```
  validates :bio, length: { maximum: 1000,
    too_long: "%{count} characters is the maximum allowed" }
end
```

This helper counts characters by default, but you can split the value in a different way using the `:tokenizer` option:

```
class Essay < ActiveRecord::Base
  validates :content, length: {
    minimum: 300,
    maximum: 400,
    tokenizer: lambda { |str| str.split(/\s+/) },
    too_short: "must have at least %{count} words",
    too_long: "must have at most %{count} words"
  }
end
```

Note that the default error messages are plural (e.g., "is too short (minimum is %{count} characters)"). For this reason, when `:minimum` is 1 you should provide a personalized message or use `presence: true` instead. When `:in` or `:within` have a lower limit of 1, you should either provide a personalized message or call `presence` prior to `length`.

2.8 `numericality`

This helper validates that your attributes have only numeric values. By default, it will match an optional sign followed by an integral or floating point number. To specify that only integral numbers are allowed set `:only_integer` to true.

If you set `:only_integer` to `true`, then it will use the
`/\A[+-]?\d+\z/`
regular expression to validate the attribute's value. Otherwise, it will try to convert the value to a number using `Float`.

Note that the regular expression above allows a trailing newline character.

```
class Player < ActiveRecord::Base
  validates :points, numericality: true
  validates :games_played, numericality: { only_integer: true }
end
```

Besides `:only_integer`, this helper also accepts the following options to add constraints to acceptable values:

- `:greater_than` - Specifies the value must be greater than the supplied value. The default error message for this option is *"must be greater than %{count}"*.
- `:greater_than_or_equal_to` - Specifies the value must be greater than or equal to the supplied value. The default error message for this option is *"must be greater than or equal to %{count}"*.
- `:equal_to` - Specifies the value must be equal to the supplied value. The default error message for this option is *"must be equal to %{count}"*.
- `:less_than` - Specifies the value must be less than the supplied value. The default error message for this option is *"must be less than %{count}"*.
- `:less_than_or_equal_to` - Specifies the value must be less than or equal the supplied value. The default error message for this option is *"must be less than or equal to %{count}"*.
- `:odd` - Specifies the value must be an odd number if set to true. The default error message for this option is *"must be odd"*.
- `:even` - Specifies the value must be an even number if set to true. The default error message for this option is *"must be even"*.

By default, `numericality` doesn't allow `nil` values. You can use `allow_nil: true` option to permit it.

The default error message is *"is not a number"*.